



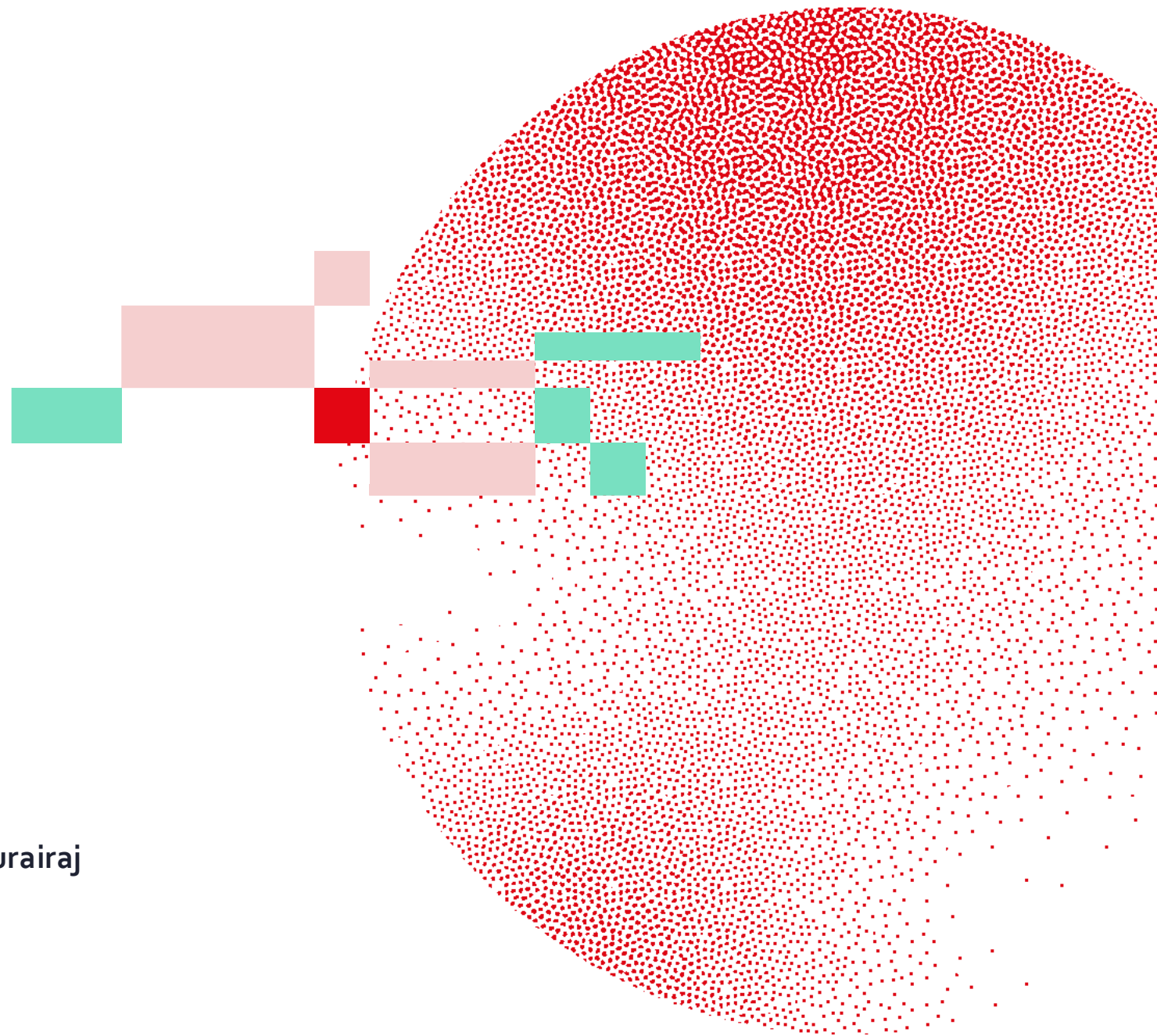
Swiss Institute of
Bioinformatics

DAY 2, PART 1

AlphaFold3

Diana Rapota, Rok Breznikar, Xavier Robin, Janani Durairaj

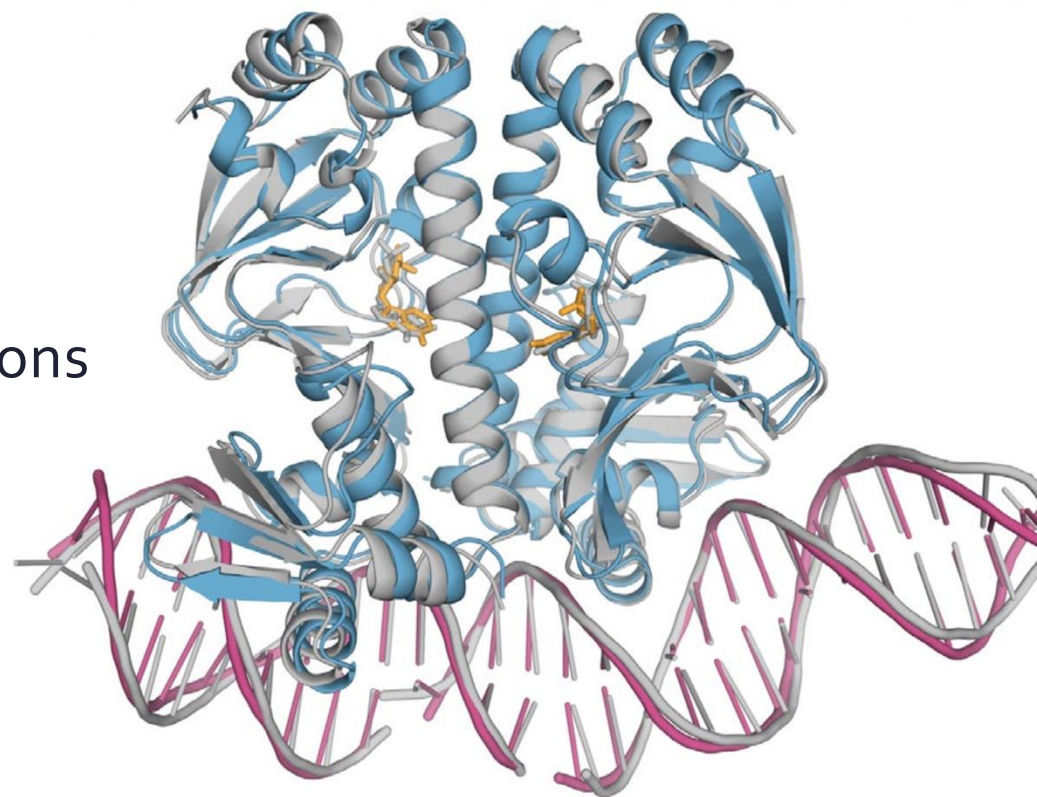
23-24 June 2026





What more you can do with it

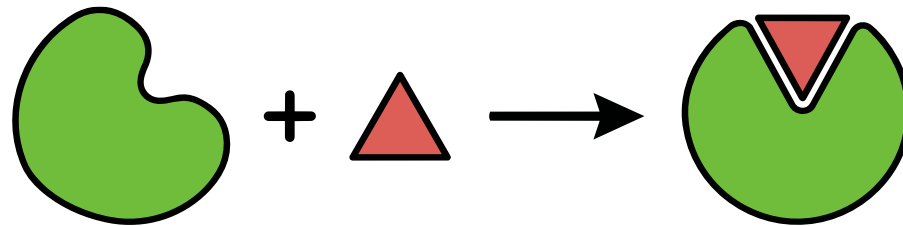
- Proteins
- DNA, RNA
- Small molecules
- Ions
- Modified residues, post-translational modifications
- Complexes of all of these
- Doesn't model:
 - Water
 - Hydrogens





Protein-ligand complex

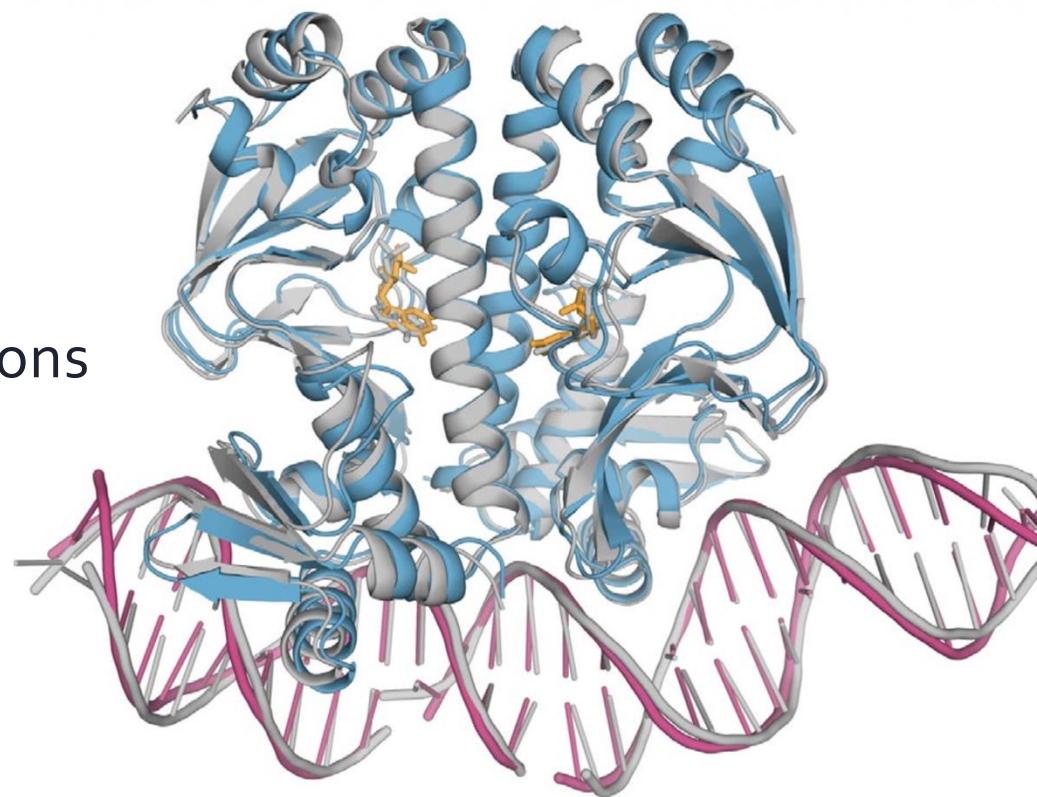
- Taking AlphaFold2 structure (protein on its own) and adding the ligand doesn't work that well (docking)
- Induced fit
(protein is not in the correct conformation for the ligand)





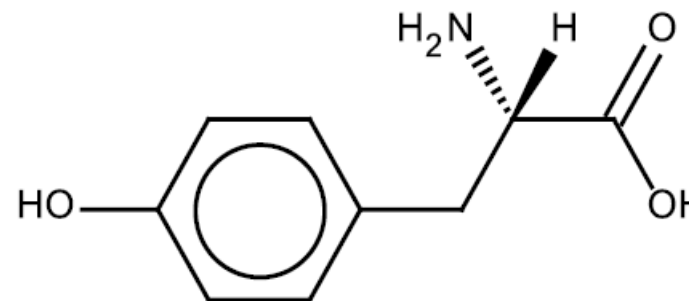
What more you can do with it

- Proteins
- DNA, RNA
- Small molecules
- Ions
- Modified residues, post-translational modifications
- Aims to model the whole complex together



SMILES

- String notation for small molecules (ligands)
- Describes a molecular graph with 4 basic rules
 - Atoms
 - Bonds
 - Branching
 - Ring closures
- Hydrogens can be included or omitted (typically omitted)





SMILES

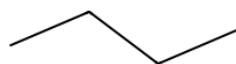
Atoms

C

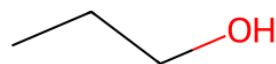
O

N

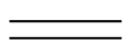
...



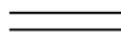
CCCC



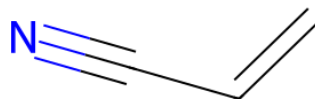
CCCO



C=C



C#C



C=CC#N

Bonds

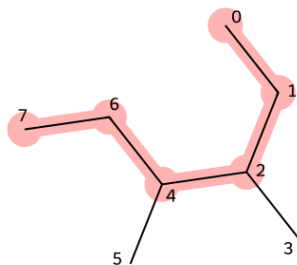
- (single)
(usually omitted)

= (double)

(triple)

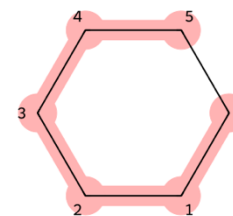
Branching

()

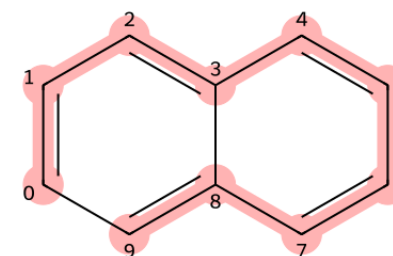


CCC(C)C(C)CC

Ring closure numbers



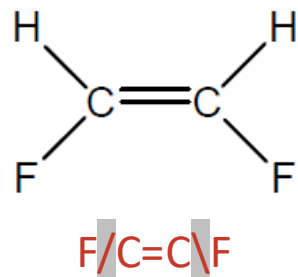
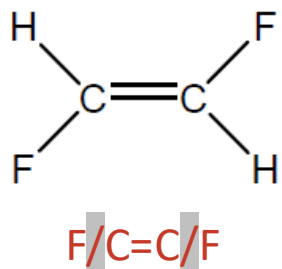
C1CCCCC1



c1ccc2ccccc2c1

Orientation around double bonds

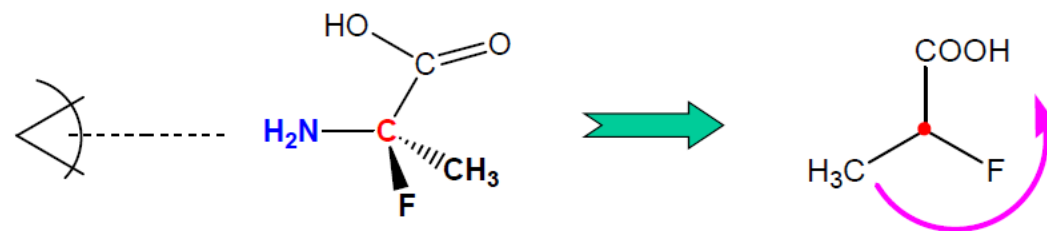
/ and \



Chiral centers

@: ↻

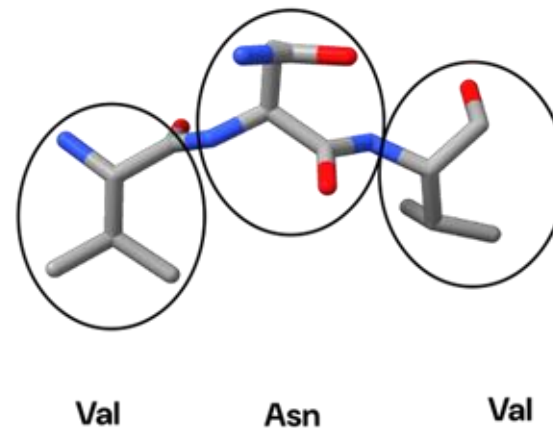
@@: ↺



N[C@](C)(F)C(=O)O

Tokenization

AlphaFold2 only dealt with proteins
→ divided into residues



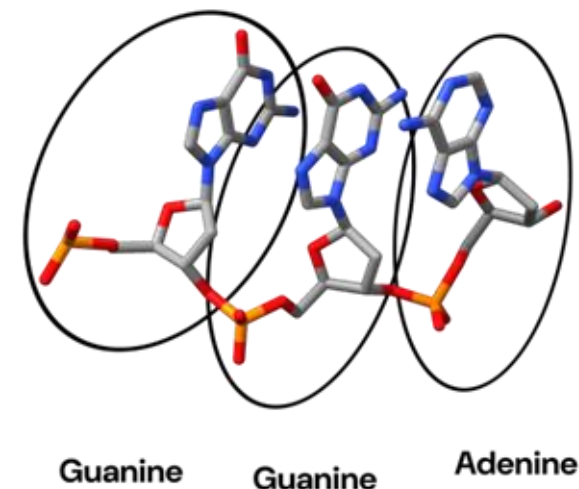
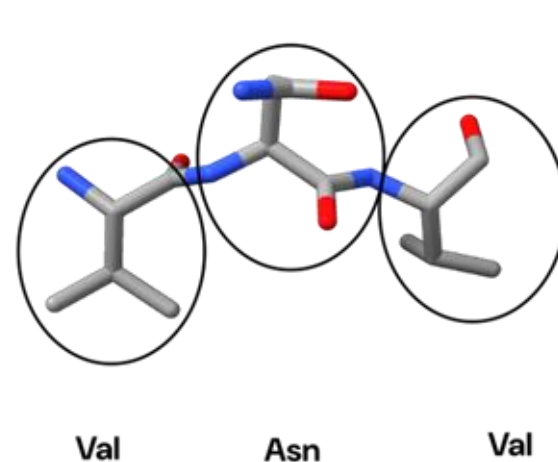


Tokenization

AlphaFold3 deals with more

Proteins → residues

DNA, RNA → nucleotides

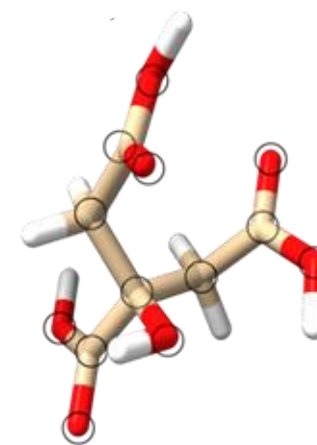


Small molecules

Ions

Modified residues

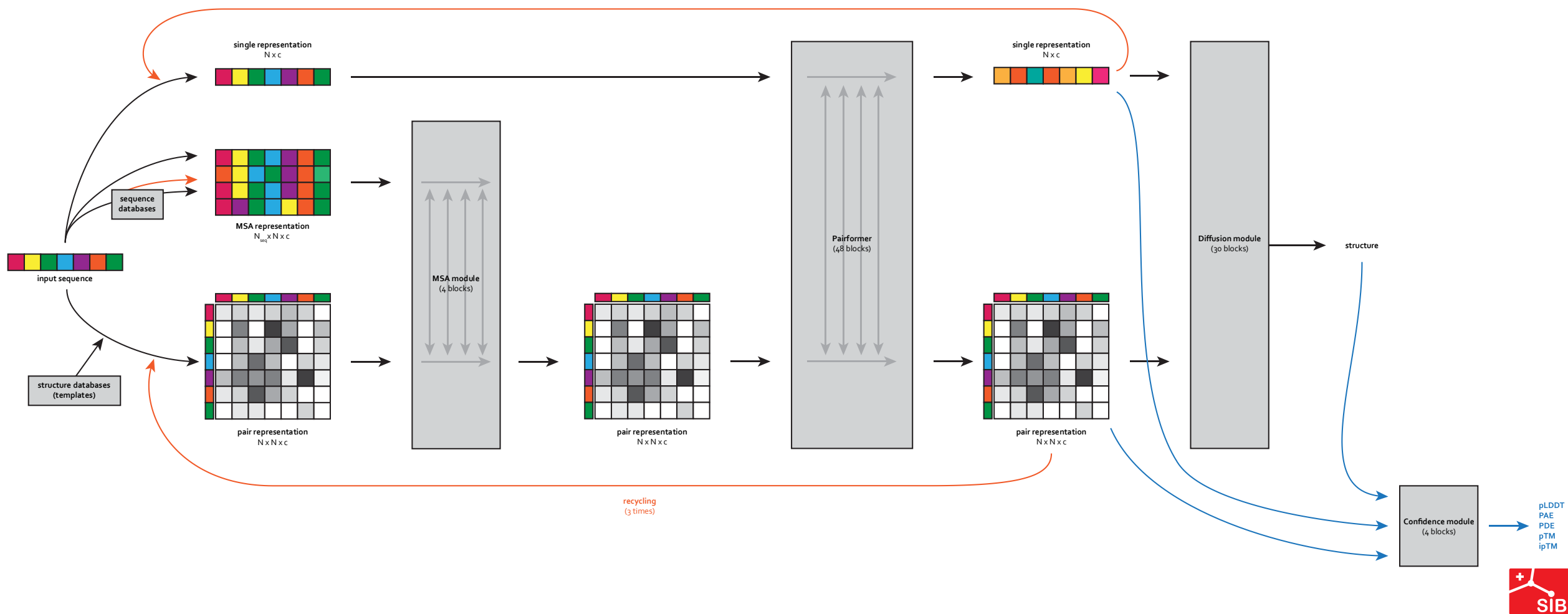
} → atoms



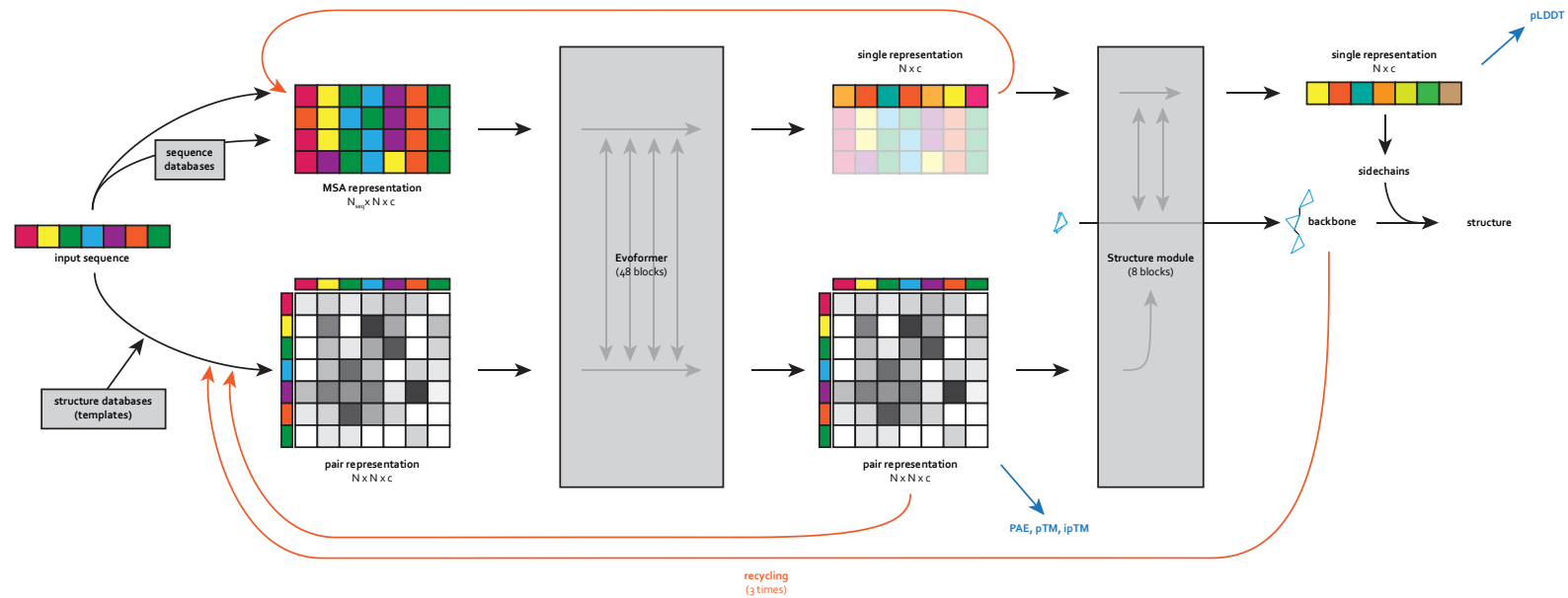
Citric Acid

Architecture

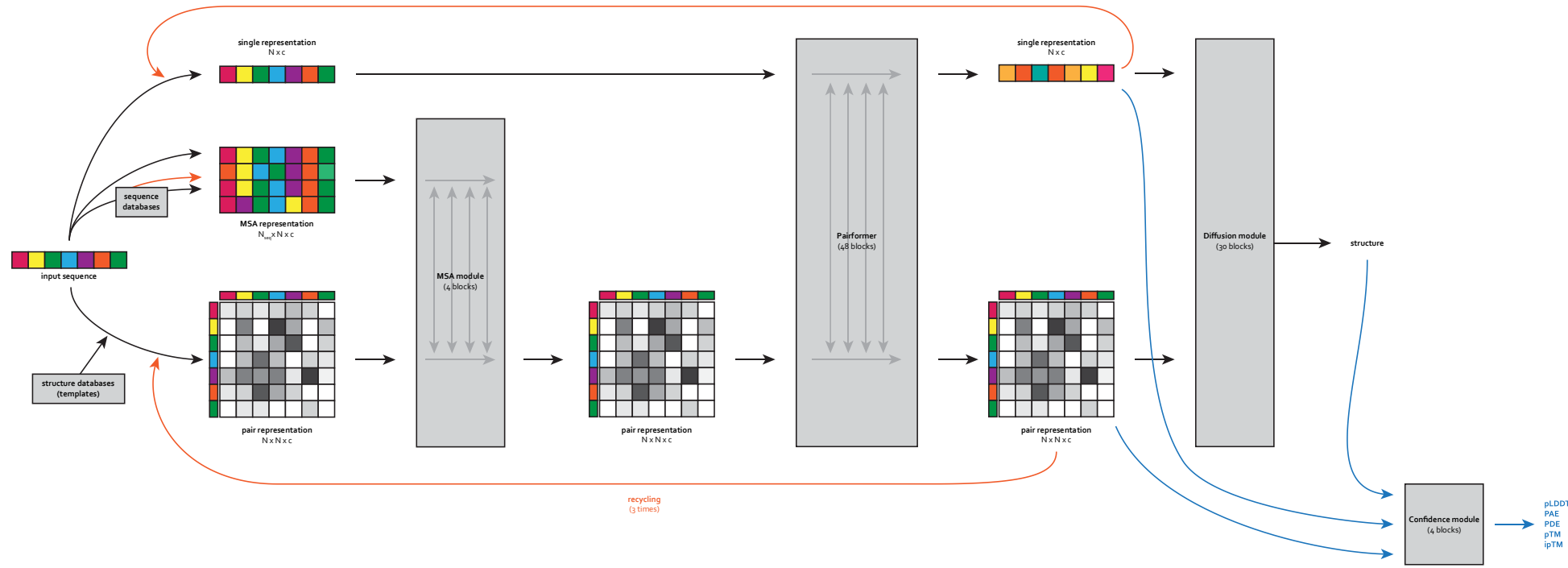
- Simplified scheme (conversions between atom and token level representations)



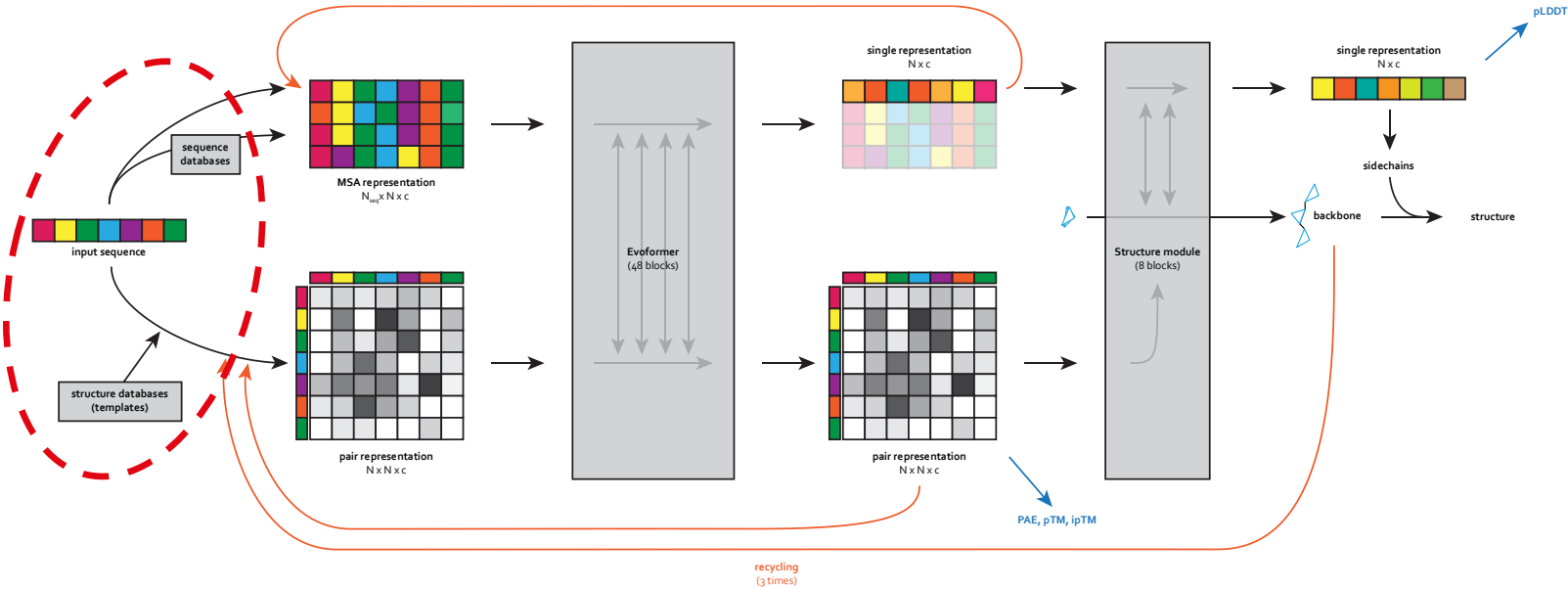
AlphaFold2



AlphaFold3

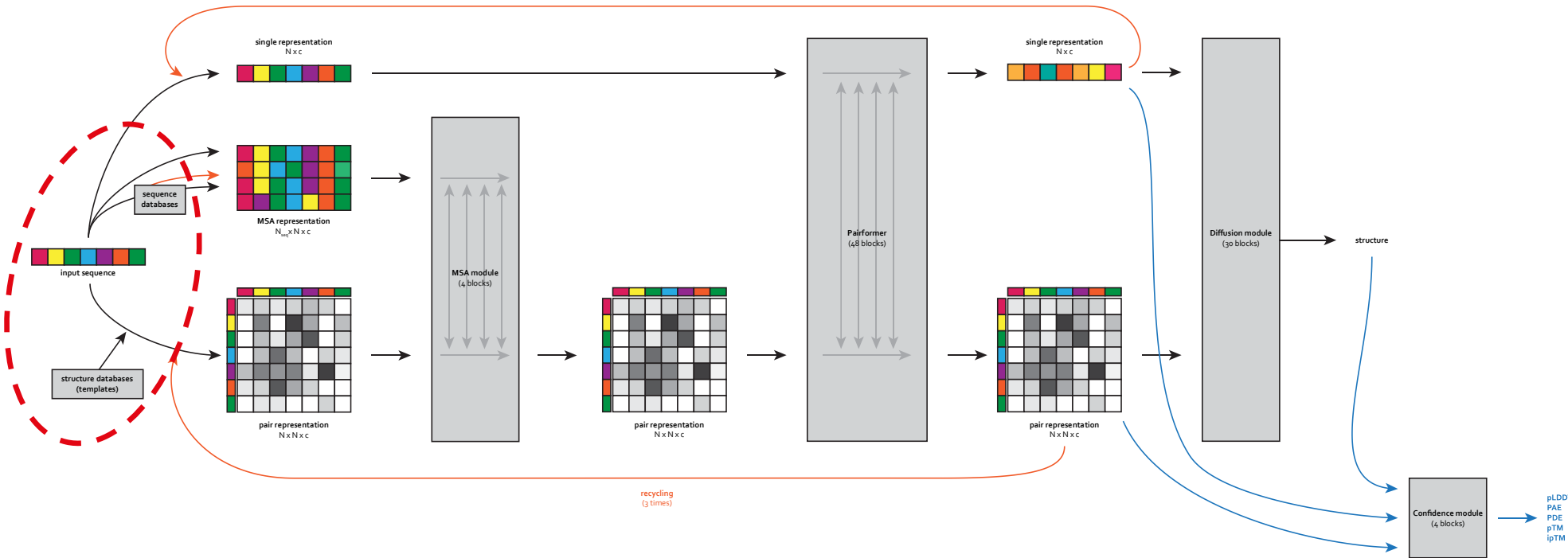


AlphaFold2

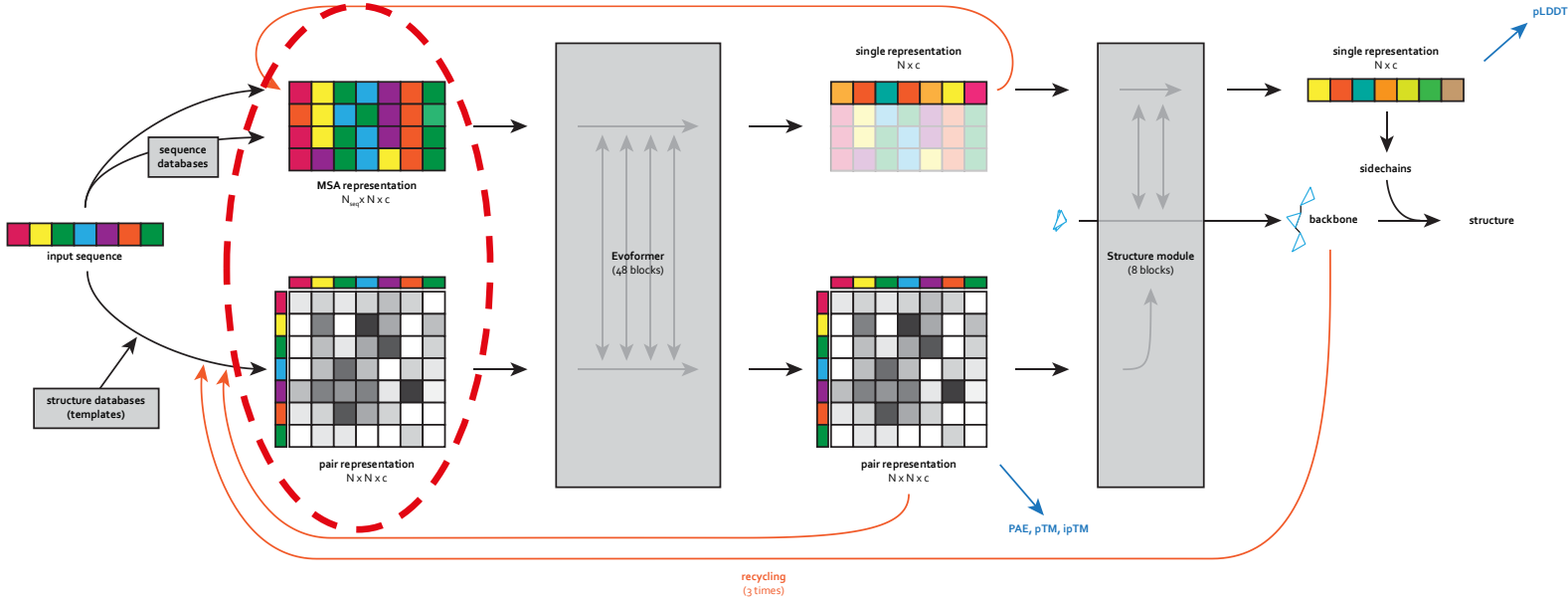


Similar inputs

AlphaFold3

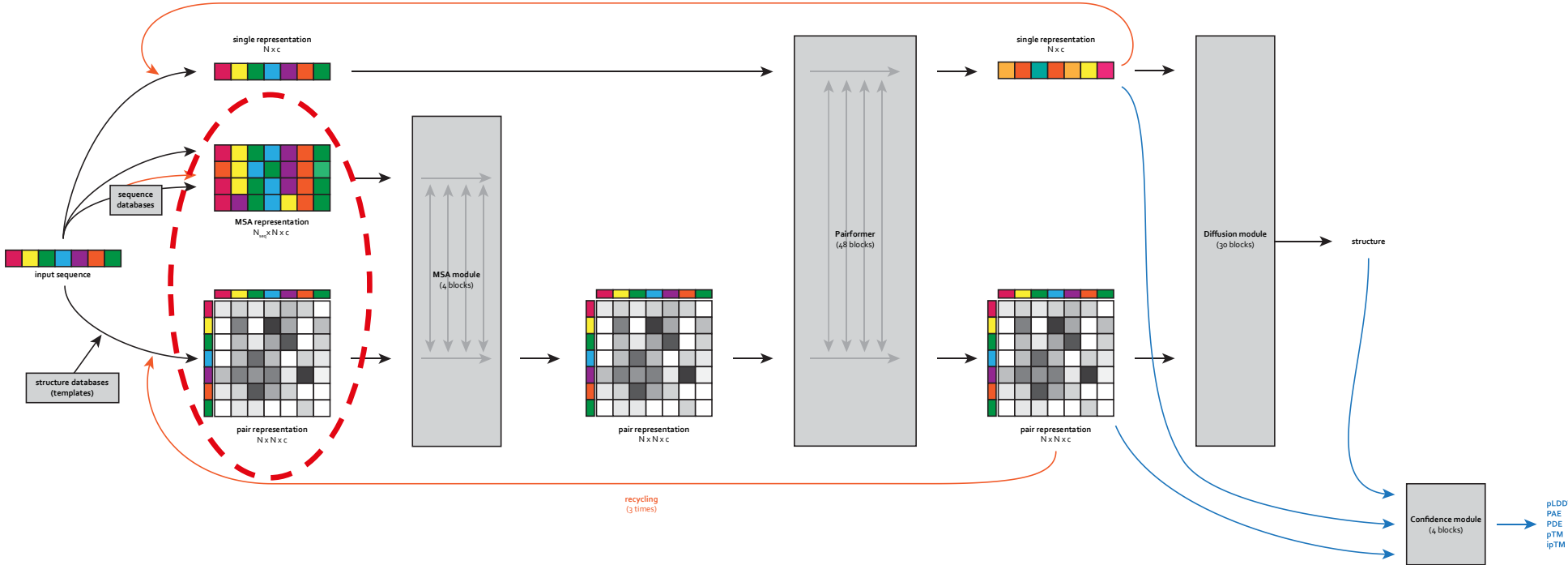


AlphaFold2

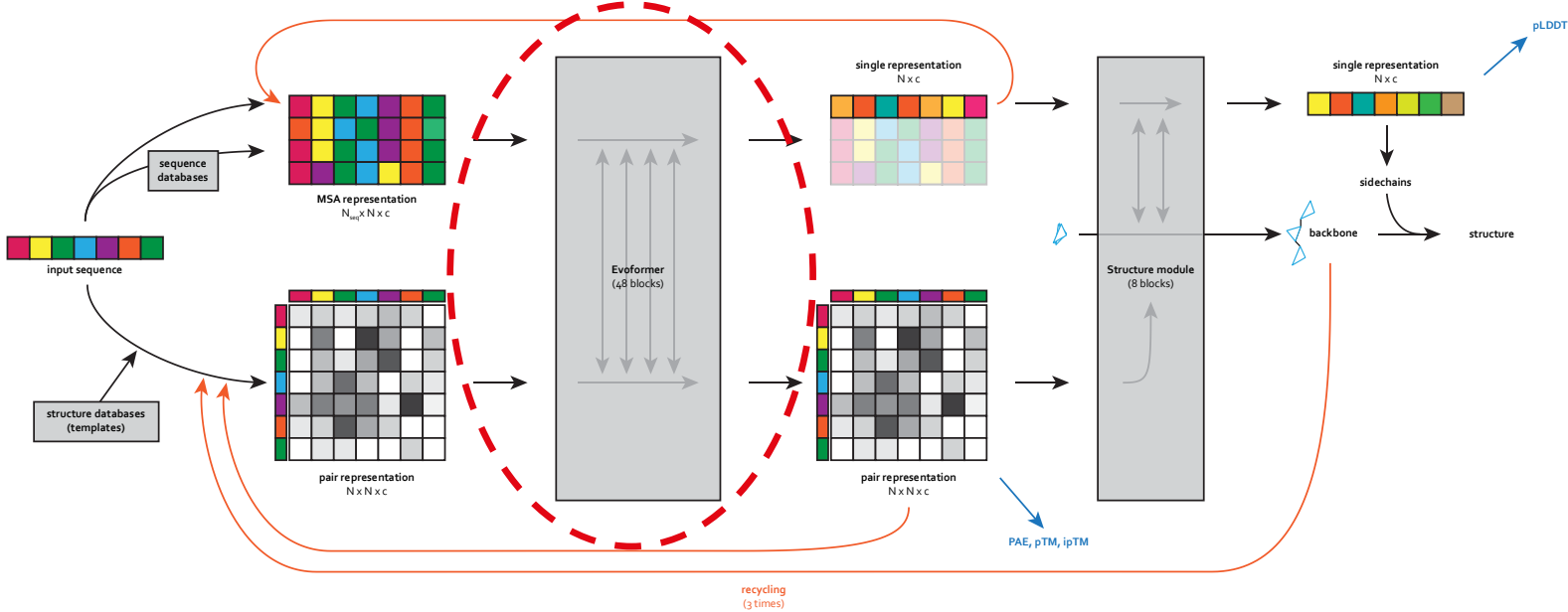


Similar representations

AlphaFold3

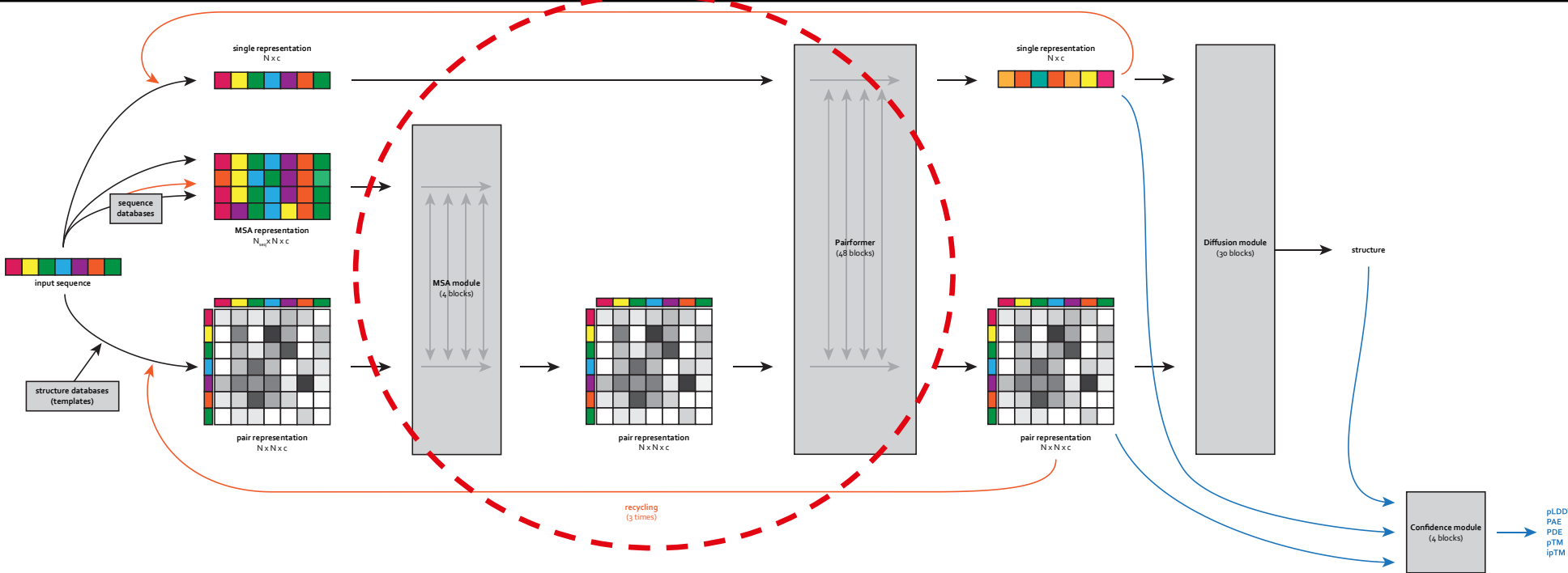


AlphaFold2

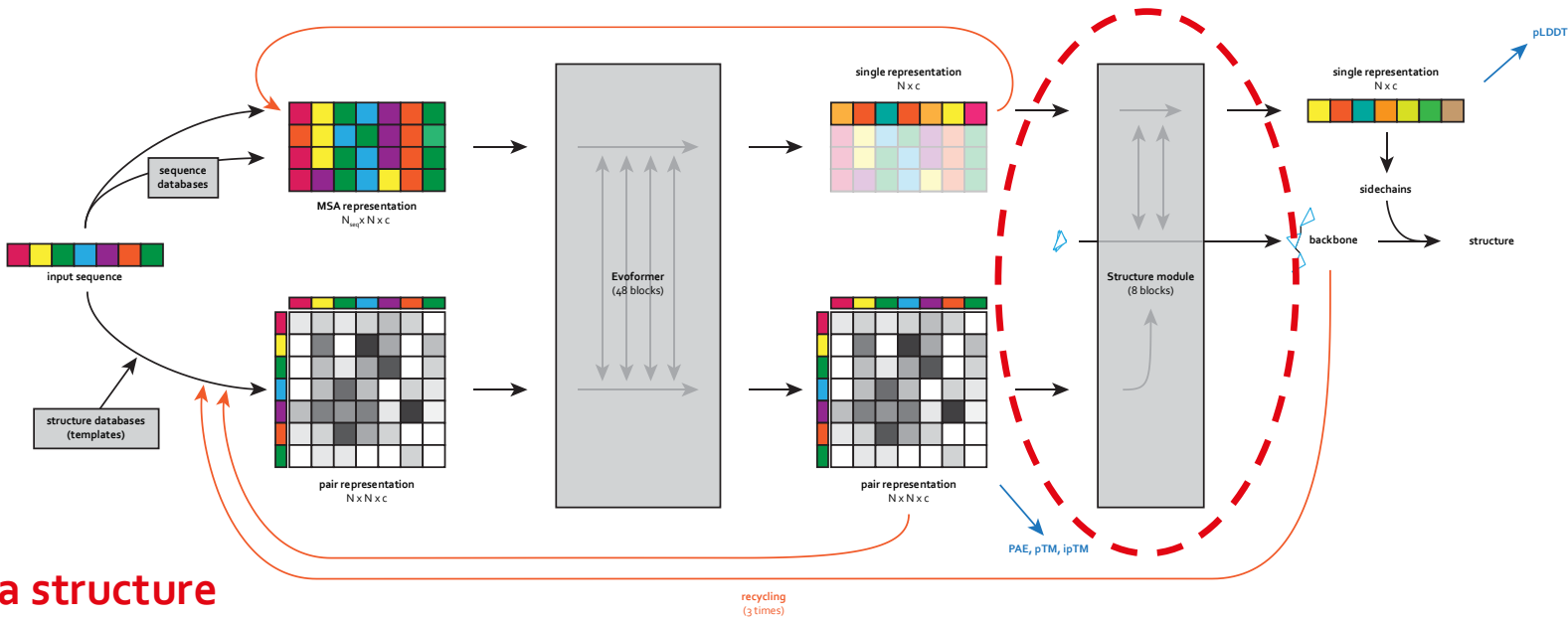


Different MSA processing

AlphaFold3

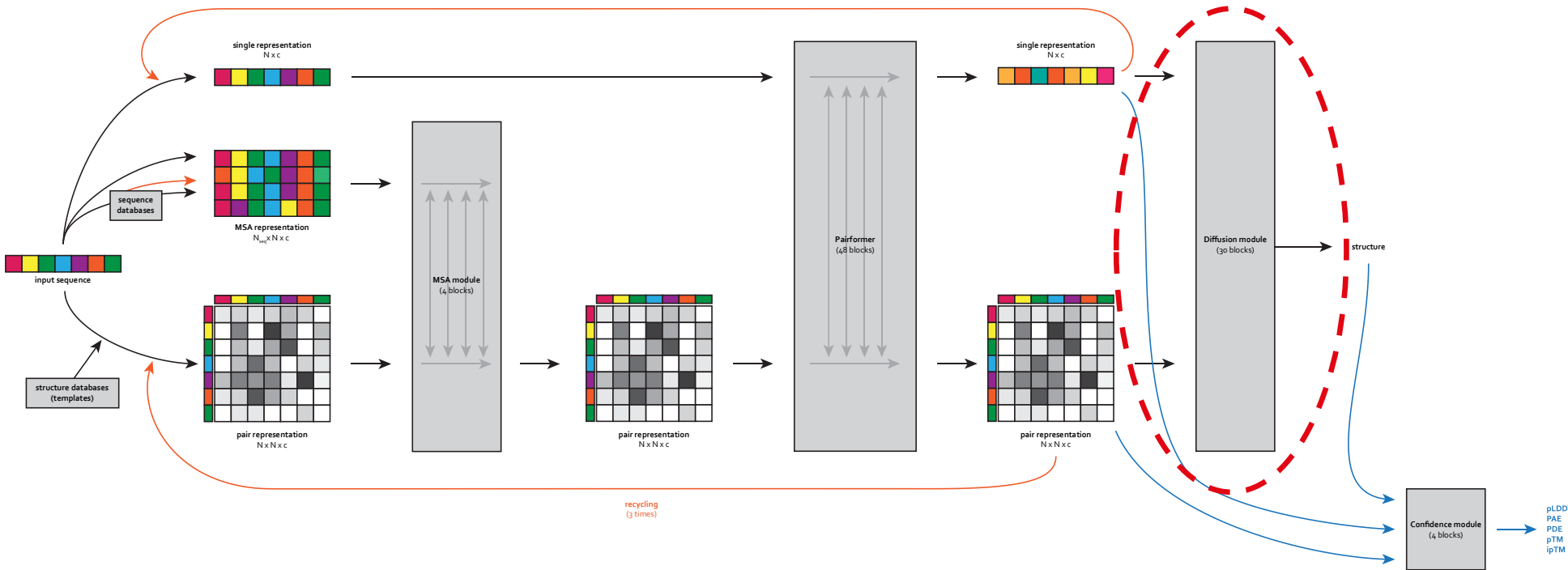


AlphaFold2



Different way of generating a structure

AlphaFold3





Structure/Diffusion module

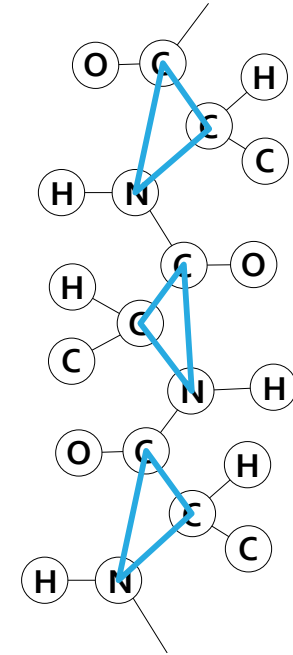
AlphaFold2

(Structure module)

- Backbone frames
- Starting at 0
- Rotations, translations



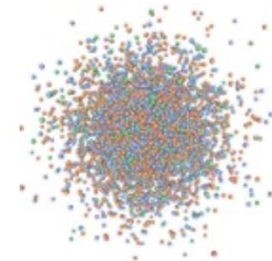
translations, rotations →



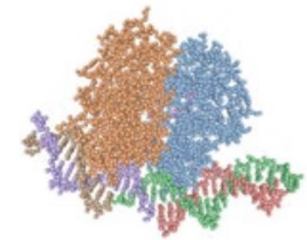
AlphaFold3

(Diffusion module)

- Coordinates of all atoms
- Starting from random coordinates
- Denoising task

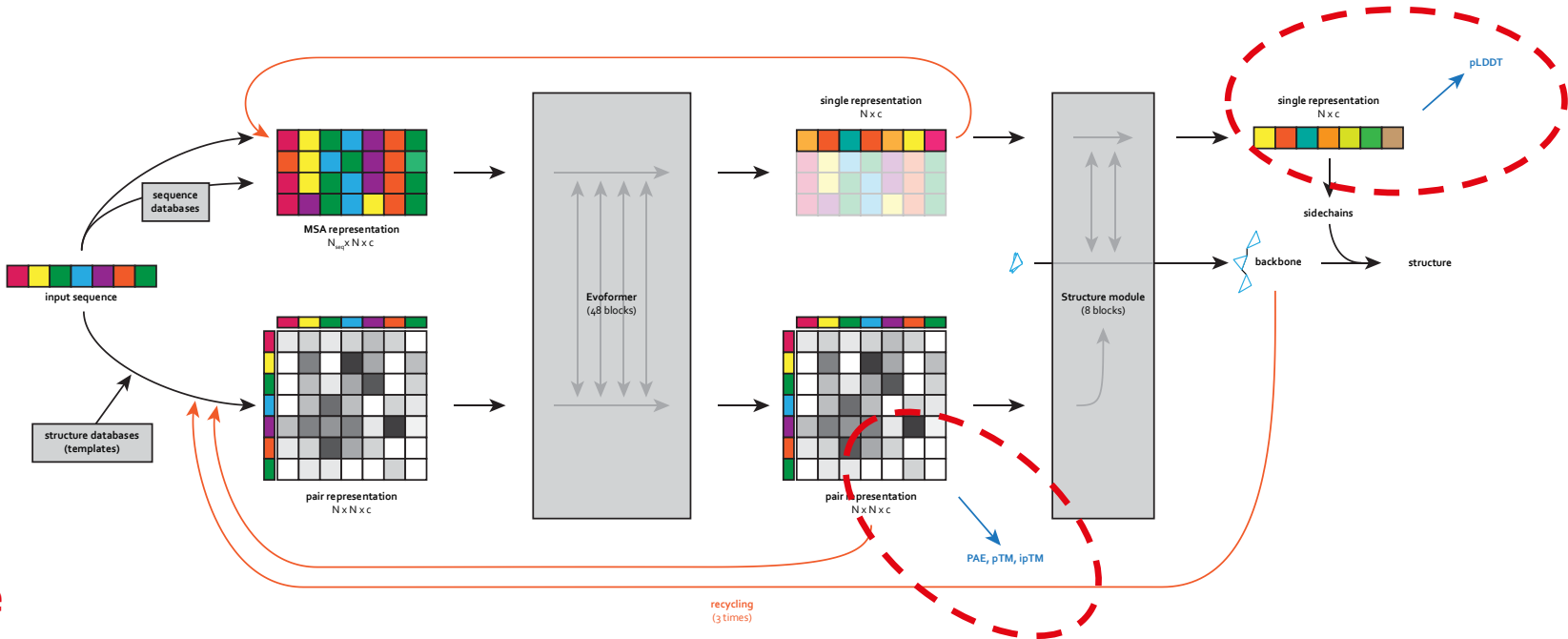


denoising →

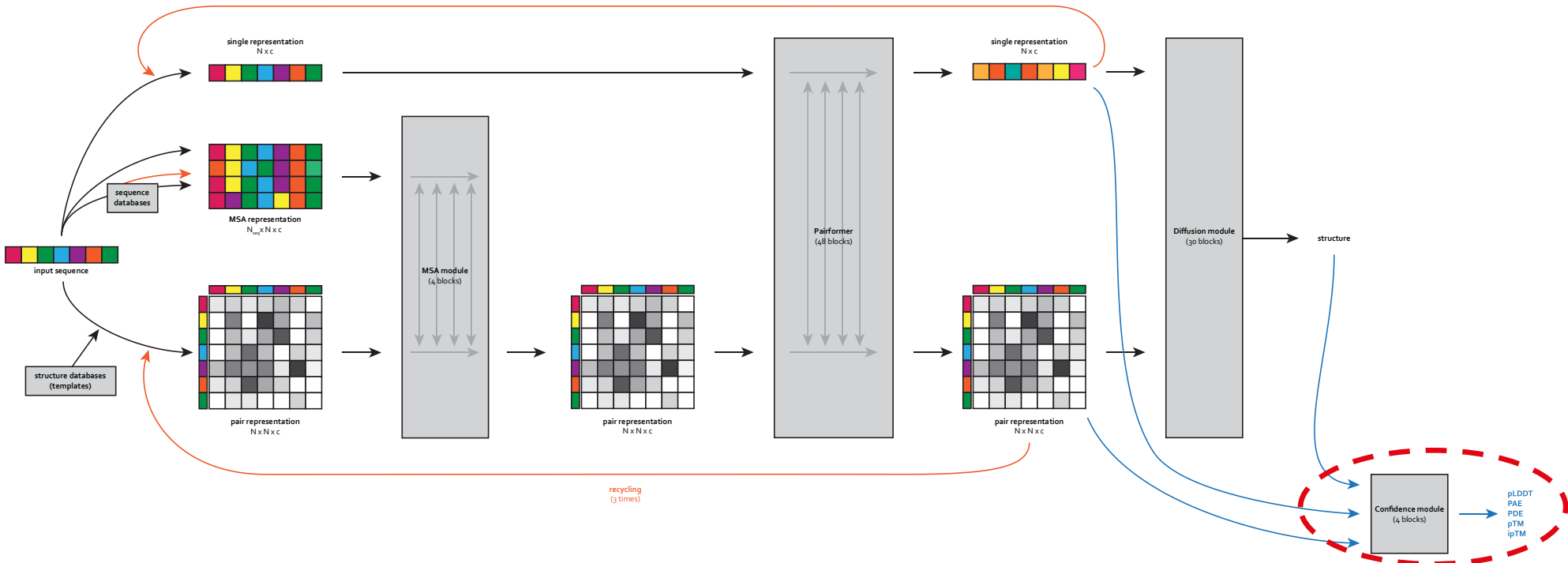


AlphaFold2

Different confidence module



AlphaFold3

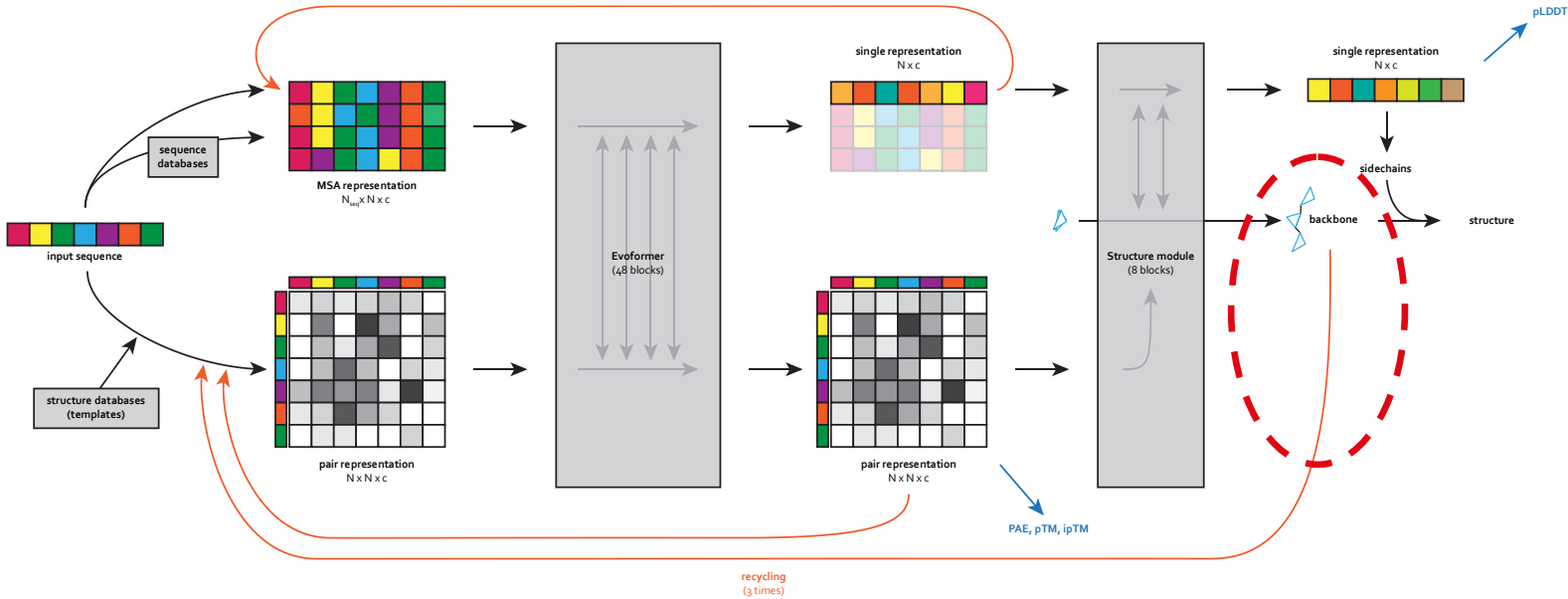




Confidence scores

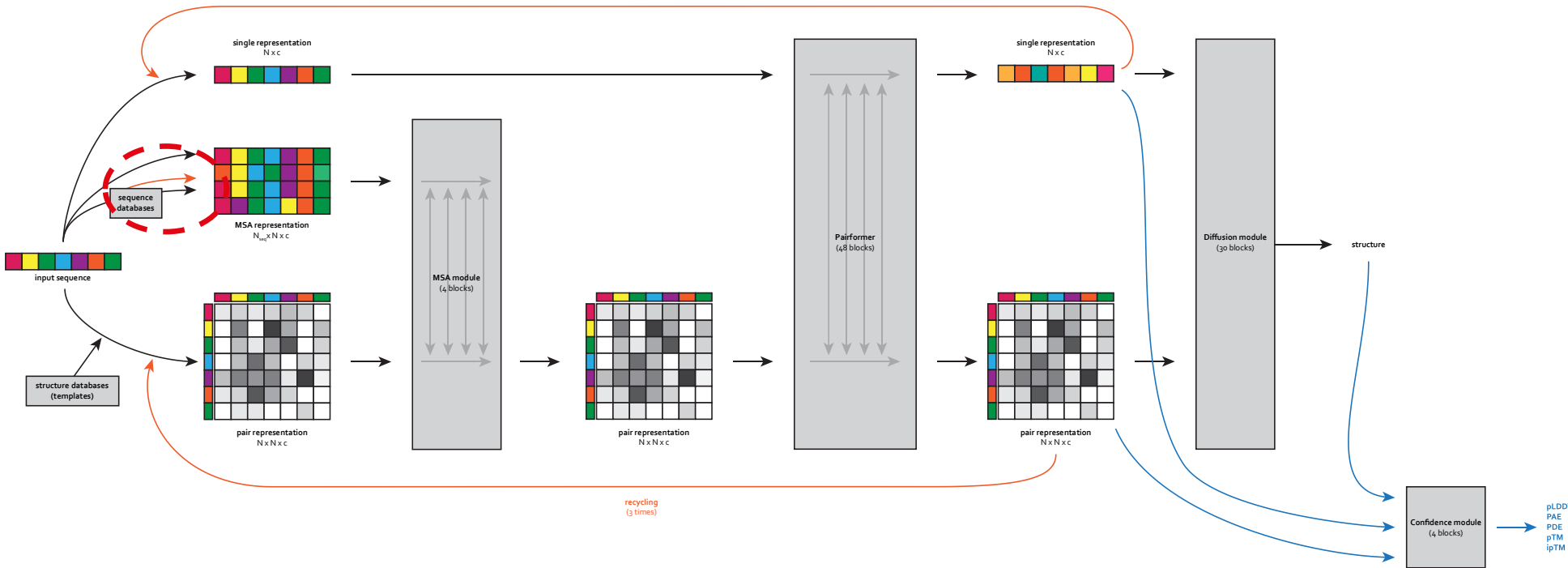
- The same metrics
- Calculated per-token
- PDE – predicted distance error
 - Predicted error in distance between two tokens (slightly different than PAE)
- Chain pair ipTM

AlphaFold2



Slightly different recycling

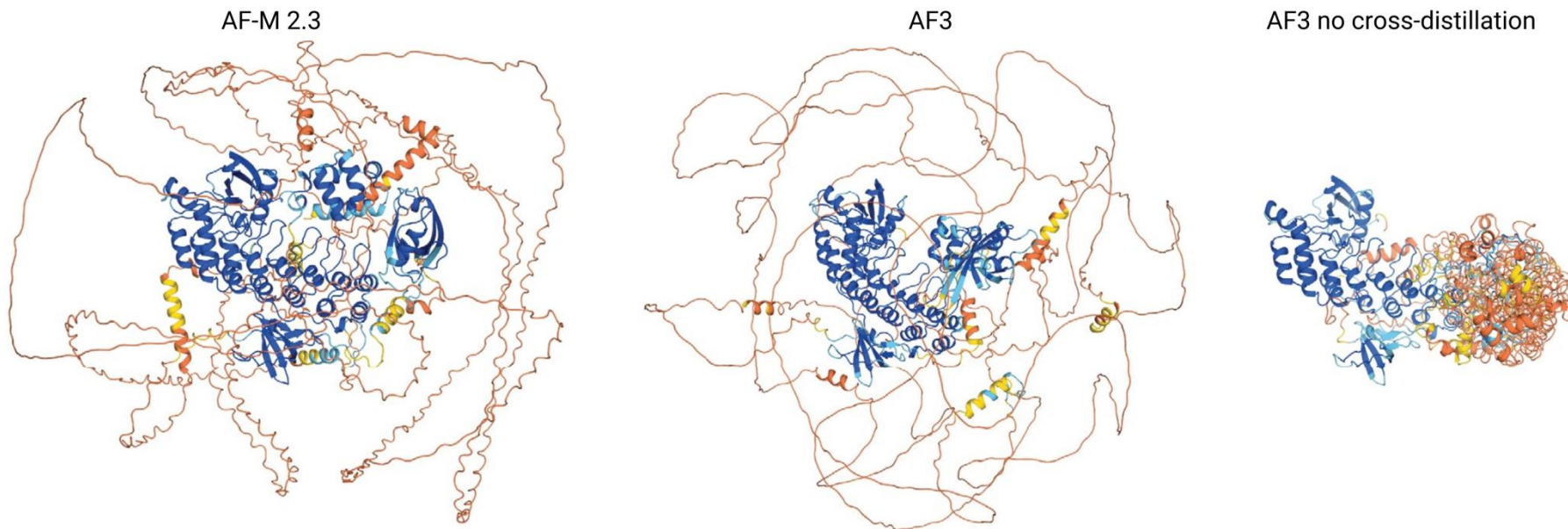
AlphaFold3





Hallucinations - structure in unstructured regions

- AlphaFold2 tends to represent these regions by long extended loops (low confidence + visual)
- AlphaFold3 might invent plausible-looking structure in these regions (still with low confidence, but people didn't like it visually)
- Cross-distillation: AF3 trained on AF2 structures to teach this behaviour

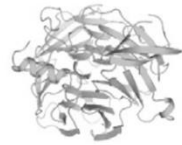




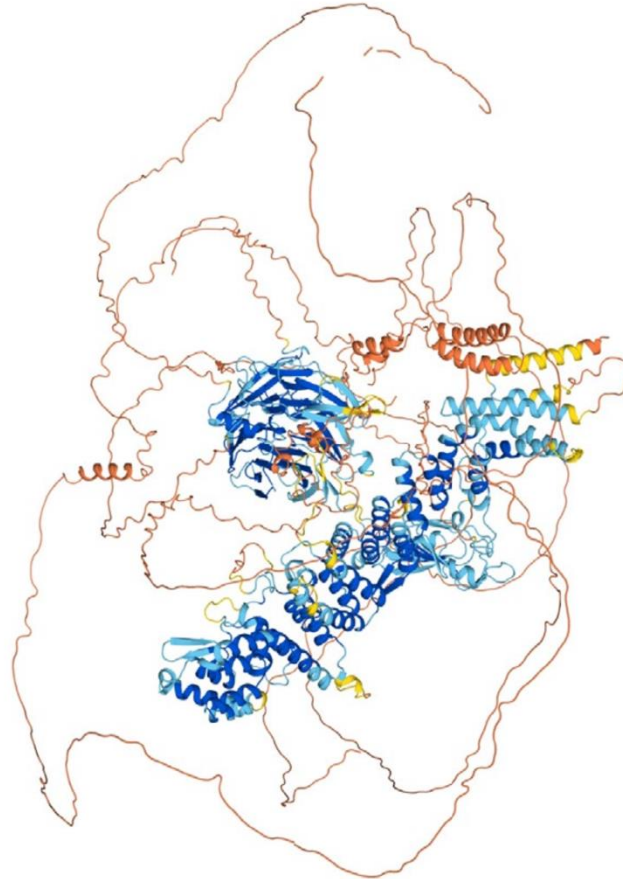
Hallucinations - structure in unstructured regions

- Hallucinations can still appear

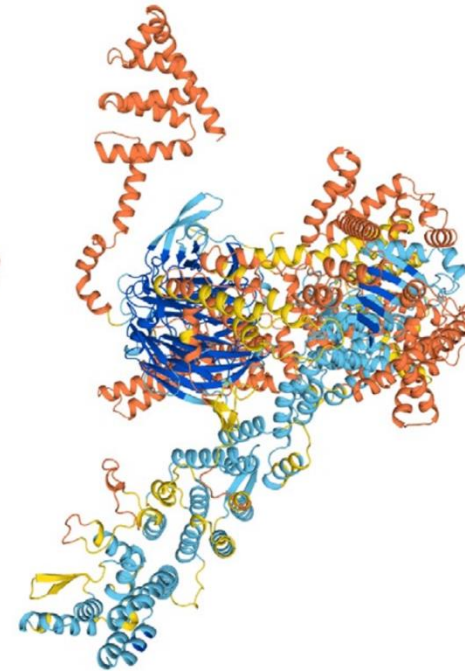
Ground truth



AF-M 2.3



AF3





How good is it?



Proteins:

similar to AlphaFold2



Protein-protein complexes:

similar to AlphaFold2



DNA:

not great, but the best for now

RNA:

there are now better tools



Protein-ligand complexes:

mostly memorizing the training data



How good is it?

Peptides:

work well

Orphan proteins:

not great (similar as AlphaFold2)

Antibodies:

work well, but require many samples (~500)

→ also check ESMFold2

Viruses:

reasonable results for host–pathogen interactions (including viruses)^[1]

→ also check FoldaVirus (for viral capsids)

Single point mutations:

not great

→ alternatives: FoldX (physics-based), ESMC (protein language model)^[2]

^[1] <https://pmc.ncbi.nlm.nih.gov/articles/PMC12258915/>

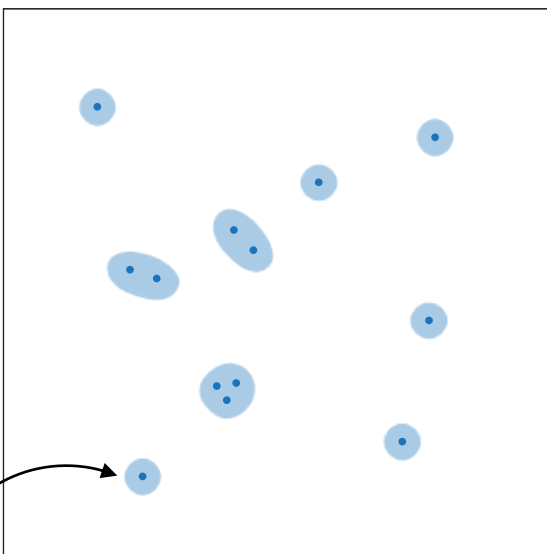
^[2] https://colab.research.google.com/github/Biohub/esm/blob/main/cookbook/tutorials/esmc_mutation_scoring.ipynb



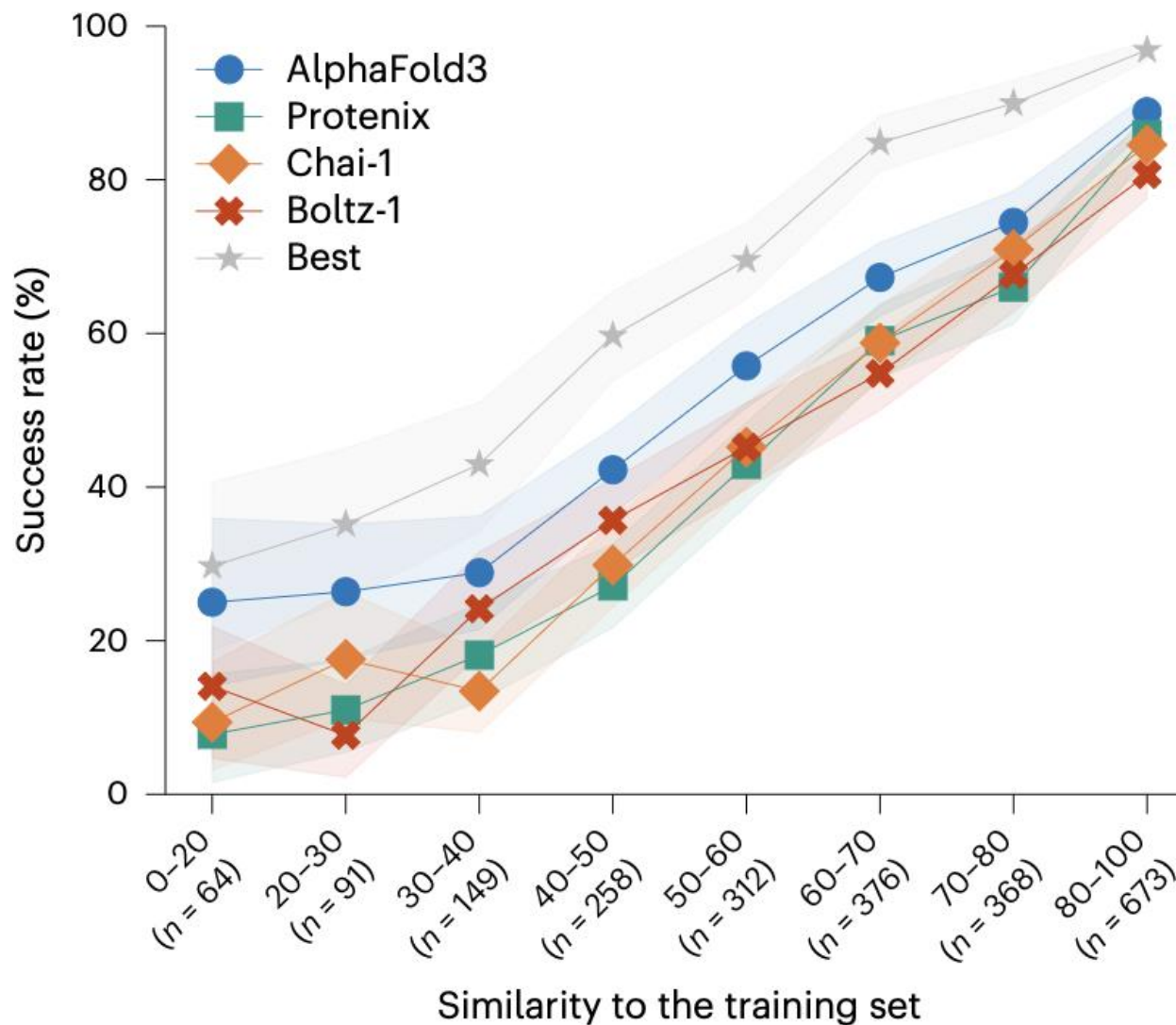
Protein-ligand complexes



- Poor performance
- No such thing as MSA for protein-ligand interaction



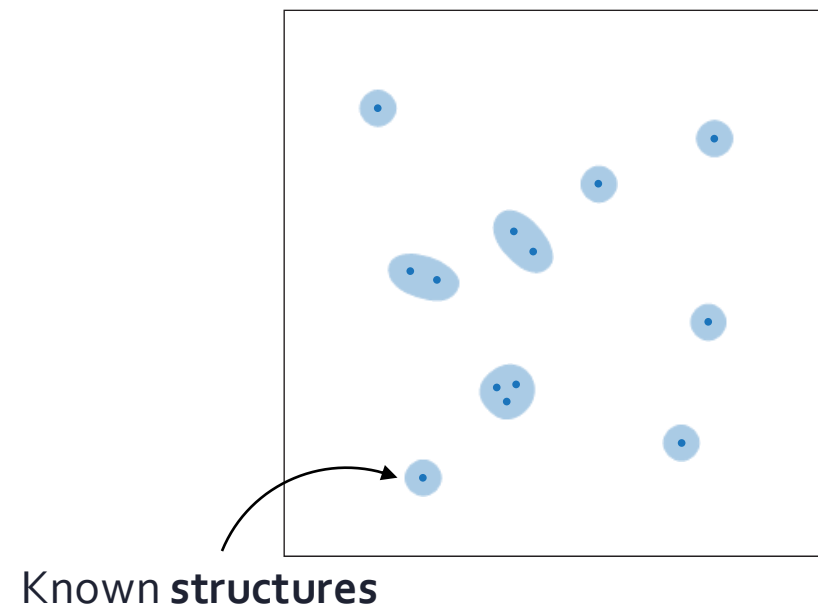
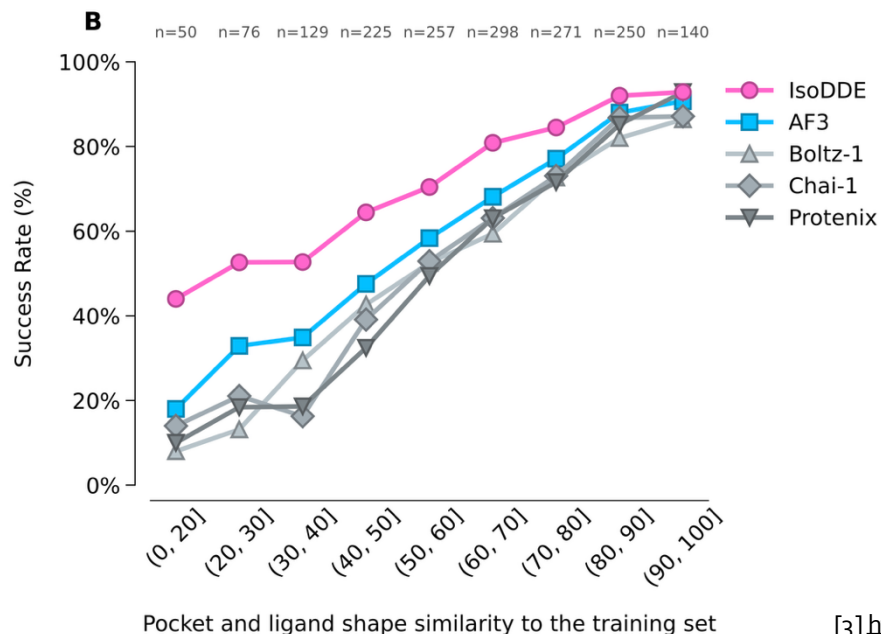
Known structures





Improving the performance on protein-ligand complexes

- More experimental structure data to train on^[1]
 - Will take years
- Synthetic data^[2]
 - Generated using physics-based methods



[1] <https://openbind.uk>

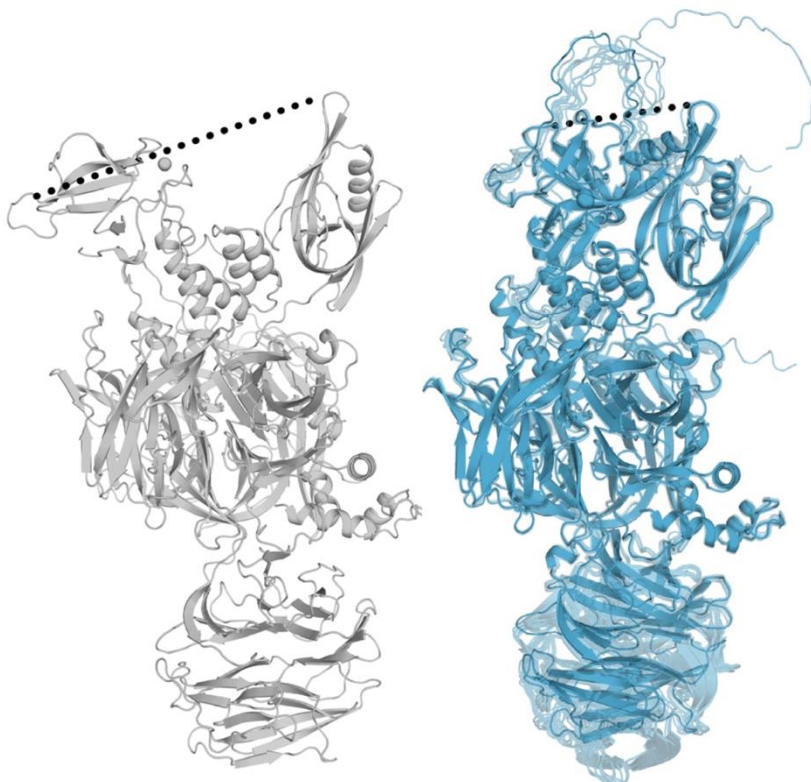
[2] <https://arxiv.org/abs/2510.24670v2>

[3] https://storage.googleapis.com/isomorphiclabs-website-public-artifacts/isodde_technical_report.pdf



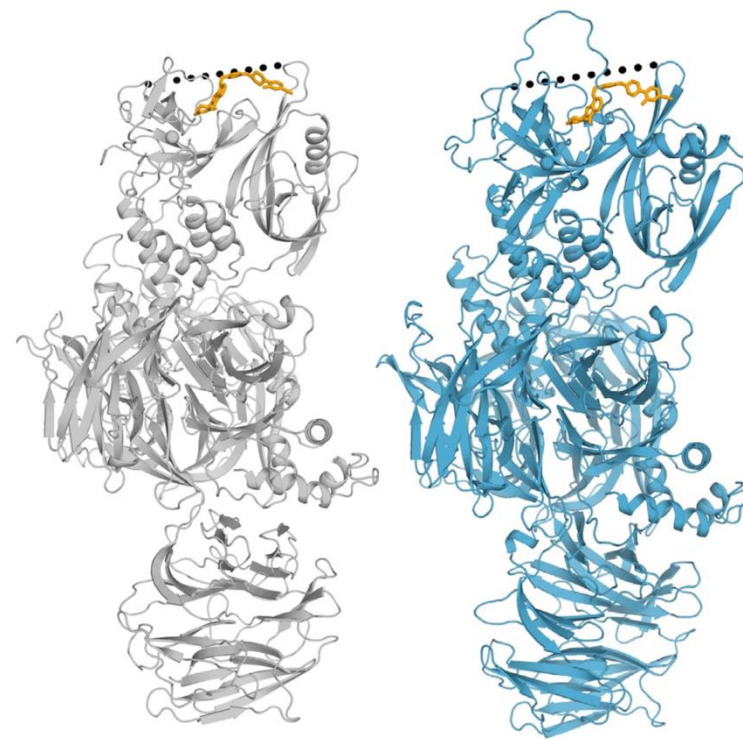
Limitations

- Conformation coverage is limited



Ground-truth
(no ligand)

AF3 prediction
(no ligand)



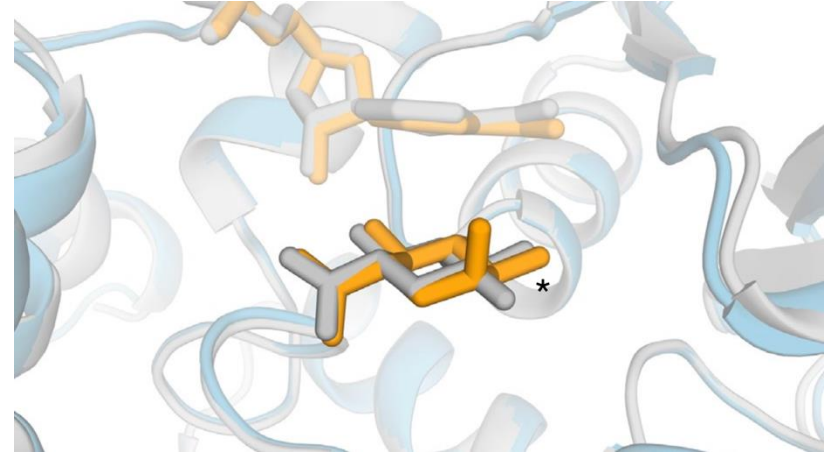
Ground-truth
(ligand-bound)

AF3 prediction
(ligand-bound)



Limitations

- Stereochemistry (chirality) violations
 - Not the one we specified as the input
- Confidence scores are overestimated
 - Users prefer AlphaFold2 metrics for some applications





Which one should I use? (AF2 or AF3)

- AF2 is freely available for both academic and commercial use
AF3 is available for non-commercial projects only
- Custom MSA/template mode is easier in AF3 compared to AF2 (for the local version)
- Online versions:
ColabFold (AlphaFold 2) allows you to adjust parameters;
the AlphaFold Server does not offer as much flexibility;
- AF2 is better for de novo proteins (due to better confidence metrics);
- Use AF3 if you want to model ligands, DNA, RNA, or Post-translational modification (PTM) – possible but not great
- AF3 is better for antibody-antigen complex prediction



How to access

- Source code
 - Hard
 - Licence limitations
- AlphaFold Server
 - Online
 - No SMILES input
- A lot of clones exist:
 - Boltz (Boltz-1, Boltz-1x, Boltz-2, Boltz-2x)
 - Chai-1 (constraints?, distance constraints, maybe from crosslinking thing, or if you know where the pocket should be (ligand-residue distance))
 - ...



How to access (AF2 and AF3)

Neurosnap



Budget Plan	Standard Plan	Academic Plan	Professional Plan
Neurosnap Budget subscription grants users access to the majority of the Neurosnap platform with an Academic License.	Neurosnap Standard subscription grants users access to the majority of the Neurosnap platform with an Academic License.	Neurosnap Academic subscription grants users access to the majority of the Neurosnap platform with an Academic License.	The Neurosnap Professional Plan grants users access to all of the Neurosnap platform with an unrestricted Commercial License.
\$6.42 per month \$76.99 billed annually	\$12.08 per month \$144.99 billed annually	\$21.25 per month \$254.99 billed annually	\$66.67 per month \$799.99 billed annually
Subscribe	Subscribe	Subscribe	Subscribe
This includes: - Academic Use 7 credits/month (monthly plans) 84 credits/year (annual plans) 7 parallel jobs - Low job priority - Most models & settings - Email Support - Guaranteed Confidentiality - Unlimited Data Storage - Full Pipelines Access - Full API Access - Max 5k Residues Protein Folding - Access to Molecular Dynamics - NeuroFold License Not Included	This includes: - Academic Use 14 credits/month (monthly plans) 168 credits/year (annual plans) 14 parallel jobs - Medium job priority - Most models & settings - Email Support - Guaranteed Confidentiality - Unlimited Data Storage - Full Pipelines Access - Full API Access - Max 5k Residues Protein Folding - Access to Molecular Dynamics - NeuroFold License Not Included	This includes: - Academic Use 25 credits/month (monthly plans) 300 credits/year (annual plans) 30 parallel jobs - Medium job priority - Most models & settings - Preferred Support - Guaranteed Confidentiality - Unlimited Data Storage - Full Pipelines Access - Full API Access - Max 5k Residues Protein Folding - Access to Molecular Dynamics - NeuroFold License Not Included	This includes: - Commercial Use 60 credits/month (monthly plans) 720 credits/year (annual plans) 60 parallel jobs - High job priority - All models & settings - Priority Support - Guaranteed Confidentiality - Unlimited Data Storage - Full Pipelines Access - Full API Access - Max 5k Residues Protein Folding - Access to Molecular Dynamics - NeuroFold License Not Included

ColabFold



Pay As You Go	Recommended Colab Pro	Colab Pro+	Colab Enterprise
EUR 10.00 for 100 Compute Units EUR 45.67 for 500 Compute Units You currently have 0 compute units. Compute units expire after 90 days. Purchase more as you need them. ✓ No subscription required. Only pay for what you use. ✓ Faster GPUs Upgrade to more powerful GPUs.	EUR 10.00 per month ✓ 100 compute units per month Compute units expire after 90 days. Purchase more as you need them. ✓ Faster GPUs Upgrade to more powerful GPUs. ✓ More memory Access our highest memory machines.	EUR 45.67 per month All of the benefits of Pro, plus: ✓ An additional 500 compute units per month, totaling 600 compute units per month. Compute units expire after 90 days. Purchase more as you need them. ✓ Faster GPUs Priority access to upgrade to more powerful premium GPUs. ✓ Background execution With compute units, your actively running notebook will continue running for up to 24hrs, even if you close your browser.	Pay for what you use ✓ Integrated Tightly integrated with Google Cloud services like BigQuery and Vertex AI. ✓ Enterprise notebook storage Replace your usage of Google Drive notebooks with GCP notebooks, stored and shared within your cloud console. ✓ Productive Generative AI powered code completion and generation.

Hardware accelerator ?

☐ CPU ☒ T4 GPU ☐ H100 GPU ☐ G4 GPU

☐ A100 GPU ☐ L4 GPU ☐ v5e-1 TPU

☐ v6e-1 TPU